

Extended Essay

Mathematics

“What is the maximum temperature a transistor can reach before it can be cooled to a safe temperature within a suitable period of time?”

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Introduction

Computers are very possibly the most influential invention of the last century. They have made several things possible which wouldn't be feasible without them like space travel, fast long-distance communication, and safe schooling opportunities during a pandemic, just to name a few. As such, it has become increasingly important as of late for people to understand how computers work as well as the components involved in letting them function.

The different parts of a computer

Computers can be broken down into several sub-components. One of these components is arguably the most important within the computer, the central processing unit (CPU). This is a chip that does all the computations and logic within the computer. It does so through the most basic building block of a computer: the transistor.

To understand how a transistor works, it is crucial to first understand the abstract concept of computer logic. Computers use a counting system called binary, which is a base-2 counting system comprised of the numbers 0 and 1. In this system, each digit represents an increasing power of 2, like how our normal base-10 counting system's digits represent different powers of 10. This base-2

system is crucially important for transistors to work because it allows us to effectively count using electricity. If we power a transistor to make it 'on', then we can equate this to a 1 in binary and an 'off' transistor would represent a zero. Using this we can then chain multiple transistors together to be able to count in binary using only transistors. This was the fundamental idea that helped create the first computers.

Creating logic with transistors

With a counting system established, we can now start to break down how transistors create logic. Using a concept known as 'logic gates', transistors can be fed into other transistors such that the output of two different transistors become the input of another transistor, which would then determine the output of that transistor. If you continue to do this, an extensive chain of transistors can create a computer based on the inputs of the lowest level of this chain.

Seeing that transistors are what causes a computer to run, it becomes clear that they are the most important component in a computer. Therefore, it is important to take their limitations into consideration. The main constraint to be

considered is heat. Transistors use electricity conducted through the semi-conducting metal silicon. Due to the nature of this metalloid [A0], as a transistor heats up or cools down, its conductivity changes.

As seen in **figure 1**, the power dissipation decreases, up to a maximum safe temperature of 134-135

degrees Celsius. If the temperature of the transistor

is above this maximum temperature however, then

if it is lowered back below that temperature again

within a period of 1 second, we assume there will

not be permanent damage to the transistor (although this temperature and time

period may vary with different types of transistors). Given these constraints, we

can now pose a question specific to this scenario:

What is the maximum temperature a transistor can reach before it can be cooled

under 134 degrees C within 1 second?

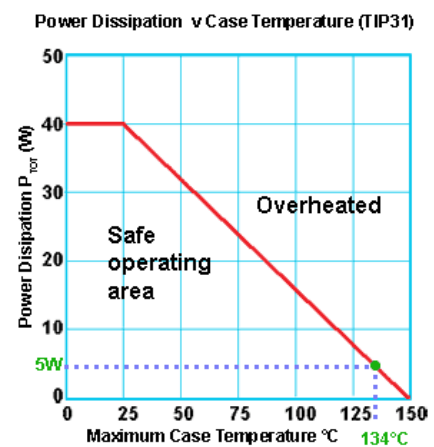


Figure 1: Graph of power dissipation vs temperature of a transistor, [A1]

Starting The Problem

In the previously posed question, we need to investigate how heat flows in a transistor over time. To do this, we can use a particular equation called the heat equation [A2], which models heat in 3 dimensions. This equation is an example of something called a partial differential equation [A3], which uses something called partial derivatives [A4] to be able to differentiate variables independently from each other by treating the other variables as constants. The heat equation is as follows:

$$\frac{\partial u}{\partial t} - c^2 \nabla^2 u = 0 \quad (1)$$

Where u is temperature and t is time.

Deriving 1D Heat Equation from 3D Heat Equation

This problem is a very difficult to solve right now. This is because of ∇ , often called Del or nabla [A5], which is a calculus operator such that:

$$\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \quad (2)$$

The equation (2) means that there are 4 different variables to work with (x, y, z , and t), making this problem extremely complex to solve in 3 dimensions.

To simplify this, we will simplify the transistor to simply a long thin bar, which is insulated on all sides. This would prevent heat loss in the y and z planes, meaning:

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial^2 u}{\partial z^2} = 0 \quad (3)$$

This results in the 3D equation being simplified to a 1D equation:

$$\frac{\partial u}{\partial t} - K \frac{\partial^2 u}{\partial x^2} = 0, \text{ or } \frac{\partial u}{\partial t} = K \frac{\partial^2 u}{\partial x^2}. \quad (4)$$

Leaving us with:

$$\frac{\partial u}{\partial t} = K \frac{\partial^2 u}{\partial x^2} \quad (5)$$

Defining Boundary Conditions and Initial Condition

In order to start to determine a solution, it is necessary that we define boundary conditions and initial conditions (in other words, what happens at each end of this hypothesized thin rod and what the initial temperature of the rod is, modelled as a function of length). For this we will use the notation $u(x, t)$, where u represents the temperature, to represent our function and the BC (boundary conditions) and IC (initial condition) will be:

$$\text{BC : } u(0, t) = u(L, t) = 0 \quad (6)$$

$$\text{IC : } u(x, 0) = f(x) \quad (7)$$

The equations of (6) mean that the temperature at both ends of the bar at every point in time are held to 0 degrees Celsius, and the equation (7) means that the temperature along the bar when no time has elapsed is equal to some function of x , which is not yet known and will be inputted when we have our general solution in order to solve for a specific scenario.

This is now an example of something called a well-defined second order linear homogenous PDE [A6], simply meaning that we now have a much simpler, comprehensible equation with which we can begin to derive a general solution and then a solution for our specific problem.

Utilizing Separation of Variables

Step one in deriving the solution for this equation is to utilize a strategy called separation of variables [A7]. This allows us to treat equation (5) as a product of two independent equations X and T , such that:

$$u(x, t) = X(x)T(t) \tag{8}$$

It is important to observe what happens with our boundary conditions here:

$$u(0, t) = X(0)T(t) = 0 \Rightarrow X(0) = 0 \tag{9}$$

$$u(L, t) = X(L)T(t) = 0 \Rightarrow X(L) = 0 \tag{10}$$

In order to separate equation (5) into the products shown in equation (8), we will have to look at the original equation and take the partial derivatives of (8) in order to reconstruct the equation (5) in terms of X and T .

$$\frac{\partial u}{\partial t} = X(x)T'(t) \quad (11)$$

$$\frac{\partial^2 u}{\partial x^2} = X''(x)T(t) \quad (12)$$

Note that since we are taking partial derivatives, we only differentiate the variable that we focus on and leave the other, treating it as a constant.

Substituting this into the 1D equation from (5), we get the new equation of:

$$X(x)T'(t) = KX''(x)T(t) \quad (13)$$

Rearranging the equation becomes:

$$\frac{T'(t)}{KT(t)} = \frac{X''(x)}{X(x)} \quad (14)$$

Since each side is now independent of the other, if t varies without x changing, the right-hand side will remain constant, meaning that both sides must be always constant. We can represent this by equating both sides to a constant, k :

$$\frac{T'(t)}{KT(t)} = \frac{X''(x)}{X(x)} = k \quad (15)$$

Rearranging this equation yet again will give us our two independent differential equations:

$$T'(t) - kKT(t) = 0 \quad (16)$$

$$X''(x) - kX(x) = 0 \quad (17)$$

Solving $X(x)$

Starting with equation (17), we need to first eliminate trivial solutions (obvious solutions which will not give much information about the problem).

There are 3 possible cases that may give trivial solutions, all to do with the value of the constant k . These cases are as follows:

$$k < 0, k = 0, \text{ and } k > 0. \quad (18)$$

Case 1: $k > 0$. We will start this by substituting k for a different constant, λ^2 .

The square on λ ensures that it is positive, meaning we will now solve for a positive constant. Doing this we get:

$$X''(x) - \lambda^2 X(x) = 0 \quad (19)$$

The solution of which is:

$$X(x) = Ae^{\lambda x} + Be^{-\lambda x} \quad (20)$$

Which can be verified by taking the second derivative of equation (20) and subtracting the second term of equation (19) as such:

$$X'(x) = A\lambda e^{\lambda x} - B\lambda e^{-\lambda x} \Rightarrow X''(x) = A\lambda^2 e^{\lambda x} + B\lambda^2 e^{-\lambda x} = \lambda^2 X(x) \quad (21)$$

$$\therefore X''(x) - \lambda^2 X'(x) = \lambda^2 X'(x) - \lambda^2 X'(x) = 0 \quad (22)$$

Applying our boundary conditions from (6) to this equation we find:

$$X(0) = Ae^{\lambda 0} + Be^{-\lambda 0} = 0 \Rightarrow A = -B \quad (23)$$

And using the substitution found in equation (23) as well as the other boundary condition, we see:

$$X(L) = Ae^{\lambda L} - Ae^{-\lambda L} = 0 \quad (24)$$

Factoring out A and multiplying both sides by $e^{\lambda L}$, we get:

$$A(e^{2\lambda L} - 1) = 0, L > 0 \Rightarrow A = 0 \text{ \& } B = 0 \quad (25)$$

Note that since L is greater than 0, $e^{2\lambda L}$ must be greater than 1, meaning that A must be equal to 0.

This is a trivial solution since $A = 0$ and $B = 0$ meaning that for all values of x and t the temperature is 0 (meaning the bar was at equilibrium from the start), so we will disregard $k > 0$ for the solution.

Case 2: $k = 0$. Rewriting the equation from (17) we get:

$$X''(x) = 0 \quad (26)$$

Whose solution is:

$$X(x) = Ax + B \quad (27)$$

Applying boundary conditions from (6) again:

$$X(0) = 0 = 0A + B \Rightarrow B = 0 \quad (28)$$

$$X(L) = 0 = AL \Rightarrow A = 0 \quad (29)$$

Yet again we have found a trivial solution since both A and $B = 0$, meaning we must move to the 3rd case.

Case 3: $k < 0$. We will represent this using $-\lambda^2$, since squaring a number ensures it is positive (provided it is not a complex number, which it is not in this case), and with a negative in front the number is always negative. This gives us:

$$X''(x) + \lambda^2 X(x) = 0 \quad (30)$$

Equation (30) is now an example of something known as a second order homogenous ordinary differential equation [A8], which is a type of equation that has very specific solutions depending on the coefficients of the terms in the equation. This specific situation gives us:

$$X(x) = A\cos(\lambda x) + B\sin(\lambda x) \quad (31)$$

This can be checked by taking the second derivative of equation (31) and adding to second term of equation (30). This is shown as follows:

$$X'(x) = -A\lambda\sin\lambda x + B\lambda\cos\lambda x \quad (32)$$

$$\Rightarrow X''(x) = -A\lambda^2\cos\lambda x - B\lambda^2\sin\lambda x = -\lambda^2 X(x) \quad (33)$$

$$\therefore -\lambda^2 X(x) + \lambda^2 X(x) = 0 \quad (34)$$

Applying our boundary conditions, we now get:

$$X(0) = A * 1 + B * 0 = 0 \Rightarrow A = 0 \quad (35)$$

$$X(L) = B\sin(\lambda L) = 0 \Rightarrow \sin(\lambda L) = 0 \quad (36)$$

In order to have a non-trivial solution to (30), B must not equal 0, meaning that $\sin(\lambda L) = 0$. This results in:

$$\lambda L = n\pi \Rightarrow \lambda = \frac{n\pi}{L}, n \in \mathbb{N} \quad (37)$$

$$\therefore X(x) = B_n \sin\left(\frac{n\pi}{L} x\right) \quad (38)$$

Where B_n is some arbitrary, unknown constant.

Solving $T(t)$

With this, we have solved the first differential equation, presented in (17).

Next, we will solve the equation in (16), with the knowledge that $k = -\lambda^2$ and $\lambda = \frac{n\pi}{L}$. Rewriting (16) with these variables we get:

$$T'(t) + \left(\frac{n\pi}{L}\right)^2 K T(t) = 0 \quad (39)$$

This is now an example of what is called a first order homogeneous ordinary differential equation [A9], which has a specific solution as follows:

$$T(t) = C_n e^{-K\left(\frac{n^2\pi^2}{\lambda^2}\right)t} \quad (40)$$

Where C_n is some arbitrary, unknown constant. This can be verified by taking the derivative of (40) and subtracting term 2 of (39), as shown below:

$$T'(t) = -C_n K \frac{n^2\pi^2}{\lambda^2} e^{-K\left(\frac{n^2\pi^2}{\lambda^2}\right)t} = -K \frac{n^2\pi^2}{\lambda^2} T(t) \quad (41)$$

$$\therefore -K \frac{n^2\pi^2}{\lambda^2} T(t) + K \frac{n^2\pi^2}{\lambda^2} T(t) = 0 \quad (42)$$

Re-combining $X(x)$ and $T(t)$

We have now found the solution to both functions of our original equation $u(x, t)$. However, each function has a different coefficient which depends on n .

Firstly, since these coefficients are both arbitrary constants, they can be combined

into another unknown constant, A_n . This constant can be dealt with through use of the superposition principle [A10], meaning that we can find our general solution to $u(x, t)$ by summing up all values of n . This gives us:

$$u(x, t) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{L}x\right) e^{-K\frac{n^2\pi^2}{L^2}t} \quad (43)$$

Using Fourier Series to Determine Coefficients

We will now narrow down the general solution of $u(x, t)$ found in (43) by using our initial condition from (7):

$$u(x, 0) = f(x) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{L}x\right) \quad (44)$$

In order to determine what the unknown coefficient A_n is, we will use something called the Fourier series [A11], represented here by $f(x)$. To do this, we must utilize a relationship called an orthogonality relationship [A12] such that:

$$\int_0^L \sin\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{L}x\right) dx = \begin{cases} 0 & n \neq m \\ \frac{L}{2} & n = m \end{cases} \quad (45)$$

We can achieve this form with our current problem by multiplying both sides by $\sin\left(\frac{m\pi}{L}x\right)$, then taking the definite integral from 0 to L of both sides. This gives us:

$$\int_0^L f(x) \sin\left(\frac{m\pi}{L}x\right) dx = \int_0^L \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{L}x\right) dx \quad (46)$$

Since $f(x)$ is a general function still, there is no way to evaluate the LHS, so it must stay the same, meaning we need to solve the RHS:

$$\int_0^L f(x) \sin\left(\frac{m\pi}{L}x\right) dx = \sum_{n=1}^L A_n \int_0^L \sin\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{L}x\right) dx \quad (47)$$

Since we are taking an infinite sum here, there will be one value for which $n = m$, and all other values of the sum will equal 0 since $n \neq m$, which is due to the orthogonality relationship defined in (45). As a result of this, we can now refer to m as n since they are equal, and get rid of our infinite sum, so that our equation simplifies to:

$$\int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx = A_n \frac{L}{2} \quad (48)$$

$$\therefore A_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx \quad (49)$$

With (43) and (49), we now have our full general solution of the 1D heat equation:

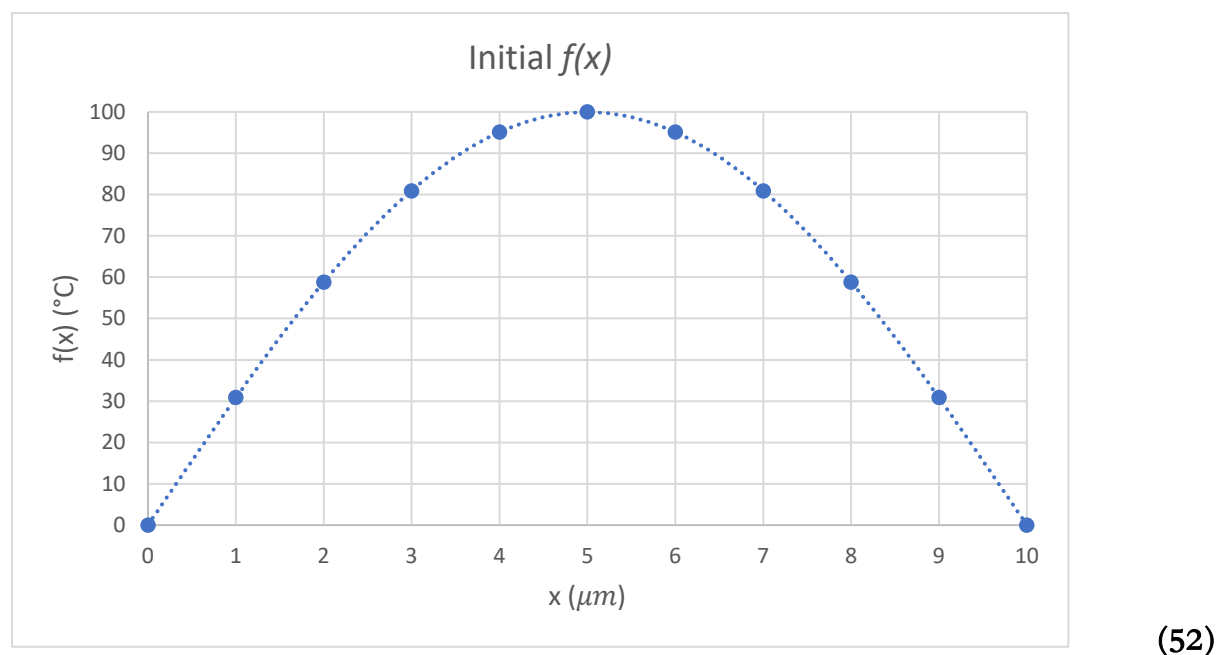
$$u(x, t) = \sum_{n=1}^{\infty} \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx \sin\left(\frac{n\pi}{L}x\right) e^{-k \frac{n^2 \pi^2}{L^2} t} \quad (50)$$

Applying Our General Solution to The Specific Problem

Let us now define the specific parameters of our transistor problem. To simulate as closely as possible a transistor that fits the defined boundary conditions in (6), we will let the initial $f(x)$ be sinusoidal with a maximum heat of X degrees

C and the bar have length $10\mu m$, which is roughly the average size of transistors used in computers today [A13]. This assumes that the heat in the transistor is highest at the central position and falls to zero at the boundaries, in the same shape as half of a sine wave. Allowing the maximum temperature to be some unknown X allows us to vary it to determine the maximum temperature for a transistor that can only support temperatures over 134 degrees C. This will cause our initial function of x to be:

$$f(x) = X \sin\left(\frac{\pi x}{10}\right) \quad (51)$$



(Model of IC with an example maximum temperature of 100 degrees C.)

The transistor in our problem is made from silicon, which has a density of 2.33 g/cm^3 , specific heat capacity of $0.71 \text{ J/g}^{\circ}C$, and thermal conductivity of 1.3

W/cm°C. This information is used to compute K , the coefficient of our exponent in (50). Now that we have defined all the necessary information for our problem, we can now check with our initial condition of $u(x, 0) = X \sin(\frac{\pi x}{10})$ to find what the expanded value of our formula is:

$$u(x, 0) = \sum_{n=1}^{\infty} A_n \sin(\frac{n\pi}{10} x) = X \sin(\frac{\pi x}{10}) \quad (53)$$

Expanding this for the first 3 terms, we see:

$$X \sin(\frac{\pi x}{10}) = A_1 \sin(\frac{\pi x}{10}) + A_2 \sin(\frac{2\pi x}{10}) + A_3 \sin(\frac{3\pi x}{10}) \quad (54)$$

What we can notice from this 3-term expansion is that the factor from our initial condition of $\sin(\frac{\pi x}{10})$ is in our first term and in no other terms, meaning that A_1 must equal X and each other coefficient must equal zero. This happens because we defined our Fourier series function to simply be a sine wave, meaning we do not need the additional terms that we would need if our Fourier series function were more complicated. By being able to utilize the first value of our coefficient A , we now eliminate our sum from (53) and take the value of n to be 1. Utilizing this, we can now write our solution for this scenario:

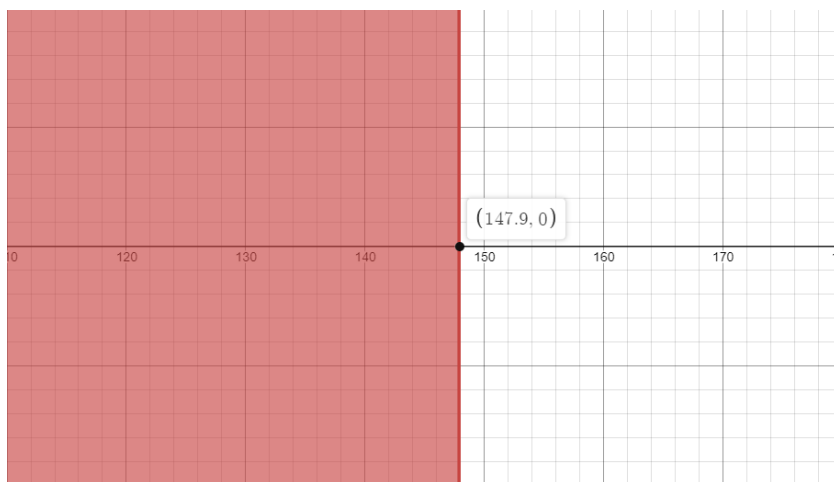
$$u(x, t) = X e^{-\frac{\pi^2}{100} t} \sin(\frac{\pi}{10} x) \quad (55)$$

Determining Maximum Temperature

Using the solution reached in (55), we can now vary the maximum temperature of our situation to find the maximum temperature before the transistor fails. We will model this by using values for t and x of 1 and 5 respectively. This will check the temperature of the middle of the transistor after one second has elapsed. This will then allow us to create an inequality to determine suitable starting temperatures:

$$134 \geq X e^{-\frac{\pi^2}{100}} \sin\left(\frac{\pi}{2}\right) \quad (56)$$

Plotting this we observe:



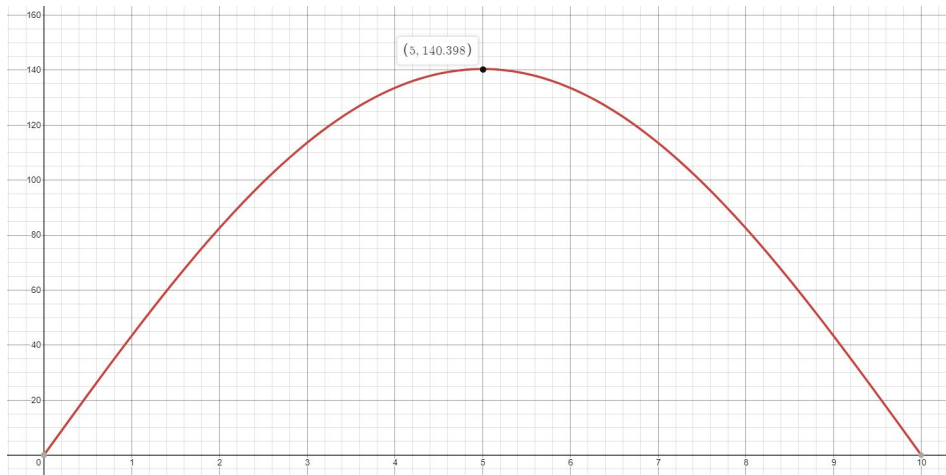
(Graph showing all suitable values of starting temperature, shown in red)

This shows that all starting temperatures in the red portion of the graph will not damage the transistor, while temperatures outside the red portion will damage

the transistor. This gives us our result, which can be achieved through the equation:

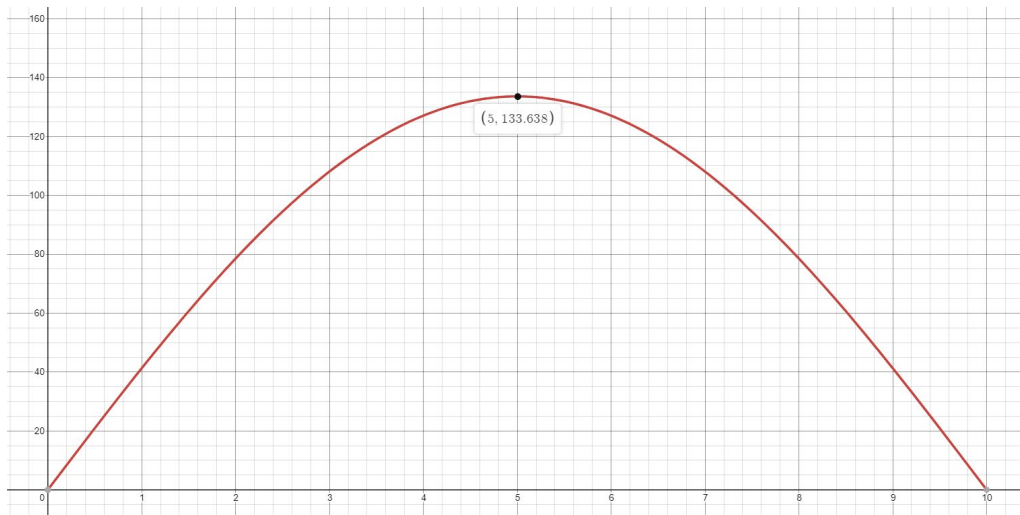
$$X = \frac{134}{e^{-\frac{\pi^2}{100} \sin(\frac{\pi}{2})}} \quad (58)$$

Through this we find that the maximum temperature that a transistor can start at to be below 134 degrees C within 1 second is 147.9 degrees C. We can verify this is true by finding the graphs for temperatures surrounding this one, such as 147.5 degrees C and 148.0 degrees C. These graphs are shown below:



(Graph of starting max temp of 147.5 at t = 0.5)

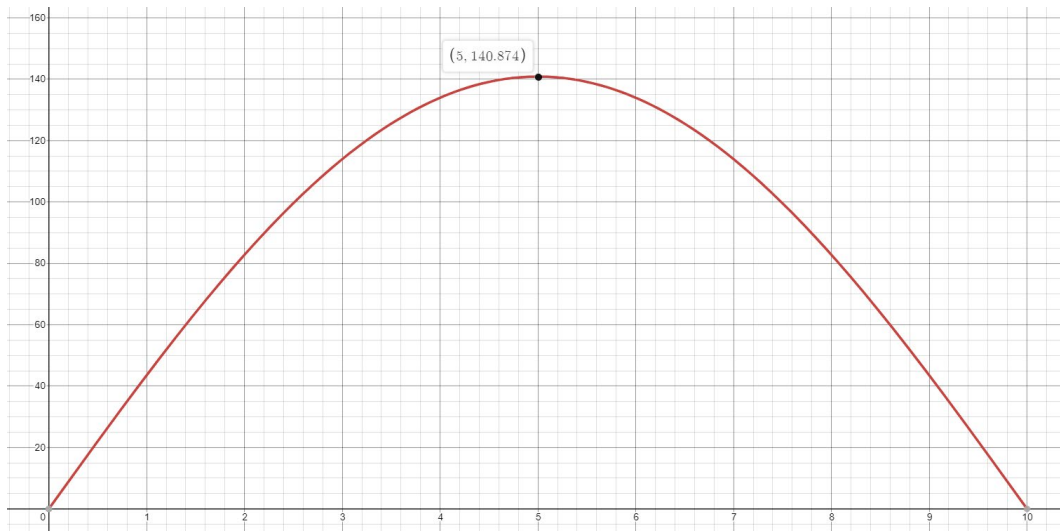
From this graph, we can see that with a starting temperature of 147.5, which has been determined to be suitable by (56) and (57), is still above the safe operating temperature after 0.5 seconds, with a temperature of 140.398 degrees C. And as expected, after 1 second has elapsed, we observe:



(Graph of starting max temp of 147.5 at $t = 1$)

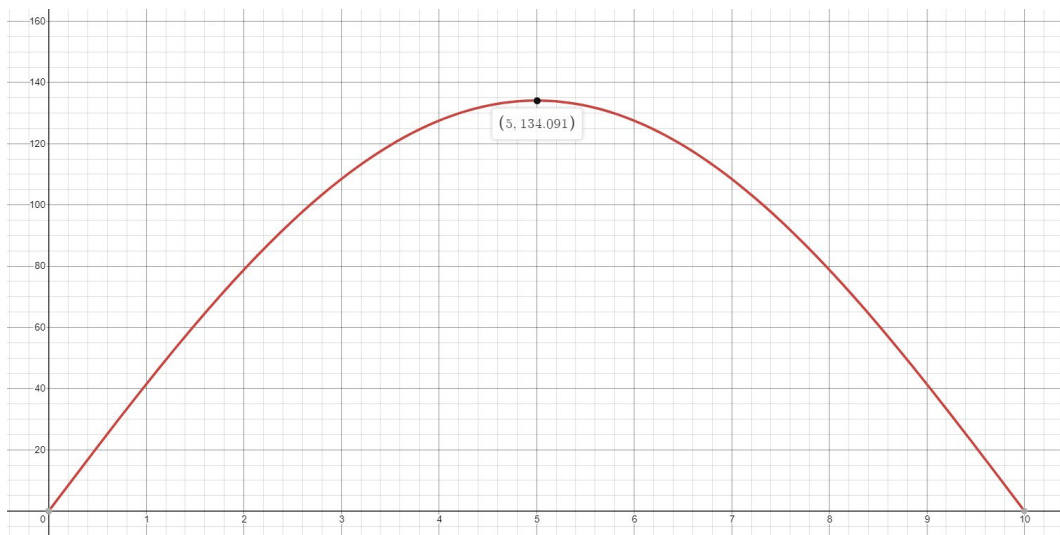
The graphs (59) and (60) show that with a starting temperature slightly lower than the solution temperature derived from (58), the final temperature after 1 second is 133.638 degrees C, which is within the suitable temperature range.

Now we will plot the graphs of a temperature slightly higher than the derived solution (47), to verify that this solution is correct. These graphs are as such:



(Graph of starting max temp of 148.0 at $t = 0.5$)

From this graph we can see that at 0.5 seconds elapsed, the temperature is only approximately 0.5 degrees C higher than the previously tested temperature from (59) at a temperature of 140.874 degrees C, however if we plot the remaining 0.5 seconds of this test we find:



(Graph of starting max temp of 148.0 at $t = 1$)

As expected, with a maximum temperature of 148.0 degrees C, the highest temperature along the bar after 1 second has elapsed is 134.091, which is just slightly above the maximum safe operating temperature of the transistor which we have modelled. As a result, we can conclusively confirm that the max temperature that the central point of a transistor defined by the BCs from (6) and the IC posed in (51) is exactly 147.9 degrees C.

Analysis

What we have now found, as mentioned before, is the maximum temperature that any point on a transistor with initial heat distributed in a sinusoidal pattern along its length in order to stay below 134 degrees C after 1 second is 147.9 degrees C. However, this is a very specific scenario. In stating this problem, we have limited the applicability of our solutions. For example, not all transistors are the size that we gave for our solution. Although average desktop computer transistors are sized at around $10\mu\text{m}$ [A12], transistors can also commonly come in sizes of 7 and $5\mu\text{m}$, and some companies like IBM and Intel have even started work toward transistors sized at $2\mu\text{m}$ and lower [A13]. This could have some impact on the suitable temperatures, as the length is a factor in the exponent that makes up equation (50). Additionally, different models of

transistors are made from slightly different materials, and have different designs to each other, allowing them to have differing maximum heat tolerances. This means that although we have found a solution for what would happen to the transistor we have imagined, it isn't exactly what would happen to other brands or models of transistor. Additionally, there may be impurities within the material of the transistor which would affect the way heat transfers throughout it, and the transistor would not realistically be a uniform piece of silicon as we have modelled it to be in this essay. These factors could lead to some inaccuracies in the model when comparing the heat dissipation of a transistor used in real life.

However, there is a different and larger consideration to be made. In this model, we are assuming that each end of the transistor is being held to zero degrees Celsius, without fail. This is problematic because it is unlikely that this would be the case in an actual computer chip since the transistors are so small and tightly packed together, which would make it improbable for each individual transistor to have temperatures held at zero at its boundaries. As a result, modelling our individual transistor with unchanging boundaries will not correctly model this situation in a real-life context. Although it is improbable for each transistor to be held to zero at each boundary however, it isn't impossible.

Modern technology and physics allow us to keep various items at temperatures of

zero degrees C and lower consistently, with relative ease, so this may not be a massive issue in a high-tech computer with an elaborate cooling system.

Another limitation to this method is that we are modelling the transistor as if it were only existing in one spatial dimension. We originally achieved this by imagining that the transistor is perfectly insulated along two of its axes, which will not be the case in a real-life example. Since transistors exist in 3 spatial dimensions and we have no feasible way to insulate each individual transistor, there will be a transfer of heat in both the y and z axes. This means to properly model this scenario, we would have to utilize the 3-dimensional heat equation, which as mentioned before equation (3), is extremely complicated.

Although there are limitations to this method, the general solution is still quite versatile, as it can account for several material types as well as different maximum safe temperatures, different starting temperatures, and different lengths of time. As a result, this is not only useful for modelling the heat transfer that occurs in a transistor but is also quite useful for modelling various other types of heat transfer.

Further Investigation

This essay has now displayed a possible way to model the heat along a transistor in a very specific scenario. However, as mentioned previously, the specificity of this scenario leads to inaccuracies within the model. Thankfully though, there are solutions to each of these inaccuracies. Firstly, the differing types of transistors can be modelled by simply using different problem parameters and different initial conditions, which would be tailored to the desired transistor. The bigger issues, however, have much more complicated solutions. To change the boundary temperatures of the hypothesized bar, one would have to change the boundary conditions of the problem, meaning the entire solution would have to be re-derived. To solve the issue of our solution not properly modelling the correct number of spatial dimensions of the transistor, our solution would have to entail solving the entire 3-dimensional heat equation, which requires advanced knowledge of partial differential equations, and would take much longer to solve than the 1-dimensional equation that we have just solved. This is because it is much more difficult to separate the 3-dimensional heat equation into independent functions like we did with our 1-dimensional equation in (8). This could be done, but it is far outside the scope of this essay and not relevant to the problem we have solved.

Disregarding issues with this model, there is a very interesting observation which could be investigated in the wake of equation (55). In this equation, we can incorporate both the maximum operating temperature and the maximum temperature that the transistor can be at before it begins to cool into our model, and create an inequality using these two variables, like we did in equation (56). It could then be investigated how the difference between the maximum operating temperature and the maximum starting temperature changes as the maximum operating temperature increases or decreases. For example, if the maximum operating temperature were to be some arbitrary value like 6 degrees C higher than in the scenario we modelled in this essay, the maximum starting temperature may change at an interesting rate.

Concluding Thoughts

With the possibilities of further investigation having been explored, we have now successfully produced an answer to our original question of what the maximum temperature a transistor can reach before it can be cooled under 134 degrees C within 1 second is. We have determined this to be precisely 147.9 degrees C, and as a result we have created a model which could be beneficial in

promoting the operating safety of what is likely the most important component of computers in the modern era.

References:

[A0] livescience.com. 2021. *Facts About Silicon*. [online] Available at:

<<https://www.livescience.com/28893-silicon.html>> [Accessed 5 October 2021].

[A1] Coates, E., 2021. *Power Transistors and Heat sinks*. [online] Learnabout-electronics.org.

Available at: <<https://learnabout-electronics.org/Amplifiers/amplifiers51.php>> [Accessed 22 September 2021].

[A2] En.wikipedia.org. 2021. *Heat equation - Wikipedia*. [online] Available at:

<https://en.wikipedia.org/wiki/Heat_equation> [Accessed 22 September 2021].

[A3] Weisstein, E., 2021. *Partial Differential Equation -- from Wolfram MathWorld*. [online]

Mathworld.wolfram.com. Available at:

<<https://mathworld.wolfram.com/PartialDifferentialEquation.html>> [Accessed 22 September 2021].

[A4] Mathsisfun.com. 2020. *Partial Derivatives*. [online] Available at:

<<https://www.mathsisfun.com/calculus/derivatives-partial.html>> [Accessed 22 September 2021].

[A5] En.wikipedia.org. 2021. *Del - Wikipedia*. [online] Available at:

<<https://en.wikipedia.org/wiki/Del>> [Accessed 22 September 2021].

[A6] Khan, S., 2010. *2nd order linear homogeneous differential equations 1 (video) | Khan Academy*.

[online] Khan Academy. Available at: <<https://www.khanacademy.org/math/differential-equations/second-order-differential-equations/linear-homogeneous-2nd-order/v/2nd-order-linear-homogeneous-differential-equations-1>> [Accessed 22 September 2021].

[A7] En.wikipedia.org. 2021. *Separation of variables - Wikipedia*. [online] Available at:

<https://en.wikipedia.org/wiki/Separation_of_variables> [Accessed 22 September 2021].

[A8] King, K., 2021. *Solving second-order homogeneous differential equations — Krista King Math |*

Online math tutor. [online] Krista King Math | Online math tutor. Available at:

<<https://www.kristakingmath.com/blog/homogeneous-differential-equations-second-order>>

[Accessed 22 September 2021].

[A9] Sfu.ca. 2021. *First Order Linear Differential Equations*. [online] Available at:

<[https://www.sfu.ca/math-](https://www.sfu.ca/math-coursenotes/Math%20158%20Course%20Notes/sec_first_order_homogeneous_linear.html#:~:text=A%20first%20order%20homogeneous%20linear,%E2%88%92p(t)y.>)

[coursenotes/Math%20158%20Course%20Notes/sec_first_order_homogeneous_linear.html#:~:te](https://www.sfu.ca/math-coursenotes/Math%20158%20Course%20Notes/sec_first_order_homogeneous_linear.html#:~:text=A%20first%20order%20homogeneous%20linear,%E2%88%92p(t)y.>)
[xt=A%20first%20order%20homogeneous%20linear,%E2%88%92p\(t\)y.>](https://www.sfu.ca/math-coursenotes/Math%20158%20Course%20Notes/sec_first_order_homogeneous_linear.html#:~:text=A%20first%20order%20homogeneous%20linear,%E2%88%92p(t)y.>) [Accessed 22 September 2021].

[A10] Borowski, E. and Borwein, J., 2007. *Collins dictionary of mathematics*. London: Collins.

[A11] Weisstein, E., 2021. *Fourier Series -- from Wolfram MathWorld*. [online]

Mathworld.wolfram.com. Available at:

<[https://mathworld.wolfram.com/FourierSeries.html#:~:text=A%20Fourier%20series%20is%20an](https://mathworld.wolfram.com/FourierSeries.html#:~:text=A%20Fourier%20series%20is%20an,the%20sine%20and%20cosine%20functions.>)
[,the%20sine%20and%20cosine%20functions.>](https://mathworld.wolfram.com/FourierSeries.html#:~:text=A%20Fourier%20series%20is%20an,the%20sine%20and%20cosine%20functions.>) [Accessed 22 September 2021].

[A12] Ocw.mit.edu. 2011. [online] Available at: <[https://ocw.mit.edu/courses/mathematics/18-](https://ocw.mit.edu/courses/mathematics/18-03sc-differential-equations-fall-2011/unit-iii-fourier-series-and-laplace-transform/fourier-series-basics/MIT18_03SCF11_s21_7text.pdf)

[03sc-differential-equations-fall-2011/unit-iii-fourier-series-and-laplace-transform/fourier-series-](https://ocw.mit.edu/courses/mathematics/18-03sc-differential-equations-fall-2011/unit-iii-fourier-series-and-laplace-transform/fourier-series-basics/MIT18_03SCF11_s21_7text.pdf)
[basics/MIT18_03SCF11_s21_7text.pdf](https://ocw.mit.edu/courses/mathematics/18-03sc-differential-equations-fall-2011/unit-iii-fourier-series-and-laplace-transform/fourier-series-basics/MIT18_03SCF11_s21_7text.pdf)> [Accessed 22 September 2021].

[A13] Thompson, A., 2016. *Scientists Have Made Transistors Smaller Than We Thought Possible*.

[online] Popular Mechanics. Available at:

<<https://www.popularmechanics.com/technology/a23353/1nm-transistor-gate>> [Accessed 22 September 2021].

[A14] Hancock, M., 2006. [online] Ocw.mit.edu. Available at:

<<https://ocw.mit.edu/courses/mathematics/18-303-linear-partial-differential-equations-fall-2006/lecture-notes/heateqni.pdf>> [Accessed 22 September 2021].

[A15] Lum, C., 2018. *Solving the 1D Heat Equation*. [online] Youtube.com. Available at:

<https://www.youtube.com/watch?v=I3jiMhVGmcg&t=2254s&ab_channel=ChristopherLum> [Accessed 22 September 2021].

[A16] Youtube.com. 2016. *PDE: Heat Equation - Separation of Variables*. [online] Available at:

<https://www.youtube.com/watch?v=VBn1diQCykQ&ab_channel=MathemaEducation> [Accessed 22 September 2021].